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[JTR]

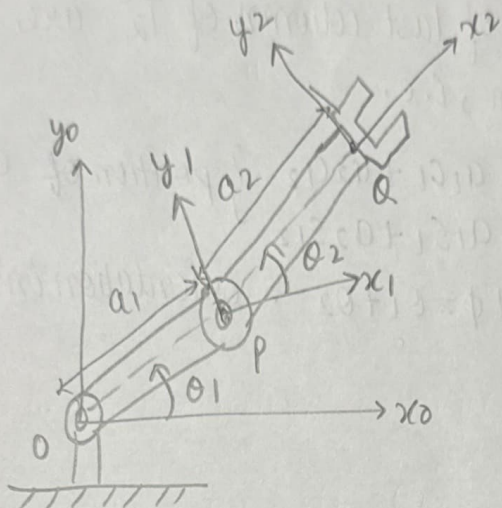
ME-639

Name - Nancy Koundal

Roll No. - 26250025

1) Pen and Paper Task →

Forward Kinematics → 2 DOF manipulator



DH Parameters for 2-link manipulator,

Link	a_i	α_i	d_i	θ_i
1	a_1	0	0	θ_1
2	a_2	0	0	θ_2

 a_i = link length d_i = link offset α_i = link twist θ_i = Joint angle

DH Transformation matrix is given by:

$$T_1^0 = A_1 \quad T_2^1 = A_2$$

$$\Rightarrow T_2^0 = T_1^0 \times T_2^1 = A_1 A_2$$

For link 1, $T_1^0 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1 c_1 \\ s_1 & c_1 & 0 & a_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ and for link 2, $T_2^1 = \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$\therefore T_2^0 = T_0^1 T_2^1 = A_1 A_2$$

$$T_2^0 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \text{where, } \begin{matrix} c_1 = \cos \theta_1 & s_1 = \sin \theta_1 & c_{12} = \cos(\theta_1 + \theta_2) \\ c_2 = \cos \theta_2 & s_2 = \sin \theta_2 & s_{12} = \sin(\theta_1 + \theta_2) \end{matrix}$$

The first two entries of last column of T_2^0 are the x and y components of the origin, i.e.

$$\begin{matrix} x = a_1 c_1 + a_2 c_{12} \\ y = a_1 s_1 + a_2 s_{12} \end{matrix} \quad \left. \vphantom{\begin{matrix} x = a_1 c_1 + a_2 c_{12} \\ y = a_1 s_1 + a_2 s_{12} \end{matrix}} \right\} \text{position of end effector}$$

and $\phi = \theta_1 + \theta_2 \rightarrow$ orientation in the plane.